

TAMPA EXECUTIVE, FL, USA

WMO: 722021

Lat: 28.014N

Lon: 82.345W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 92816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	3.0	-4.9	2.5	7.8	-2.0	3.2	9.7	8.3	21.5	7.4	20.5	0.6	20	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	35.2	25.3	34.8	25.3	33.8	25.2	27.4	32.1	27.0	31.8	26.6	31.4	1.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.7	29.4	26.0	21.3	29.2	25.2	20.3	28.6	86.2	32.0	84.0	31.9	82.5	31.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.7	5.0	-2.4	36.7	4.3	1.2	-5.4	37.5	-7.9	38.2	-10.3	38.9	-13.4	39.7		
			-2.8	28.0	3.7	0.5	-5.5	28.4	-7.7	28.7	-9.8	29.0	-12.5	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	15.5	17.5	19.4	22.2	25.3	27.7	28.5	27.6	24.1	19.4	18.0		
		DBStd	5.53	4.79	4.42	3.91	2.91	2.19	1.41	1.28	1.30	3.21	3.91	4.42		
		HDD10.0	18	11	4	1	0	0	0	0	0	0	1	2		
		HDD18.3	318	113	62	37	6	0	0	0	0	5	33	62		
		CDD10.0	4692	181	213	292	367	474	531	564	527	437	284	249		
		CDD18.3	1951	24	38	69	123	216	281	306	315	277	183	66		
		CDH23.3	18927	193	328	658	1309	2325	2861	3007	3143	2619	1576	528		
		CDH26.7	8024	23	56	162	471	1093	1389	1410	1454	1177	611	111		
(14)	Wind	WSAvg	1.6	1.9	1.9	2.0	1.9	1.7	1.3	1.0	1.2	1.4	1.5	1.6	1.7	
(15)	Precipitation	PrecAvg	1328	74	66	87	65	77	205	207	210	174	70	43	60	
		PrecMax	1665	158	234	206	146	308	455	338	335	407	169	150	325	
		PrecMin	946	16	7	4	4	11	26	101	144	60	2	3	3	
		PrecStd	186	38	50	58	33	64	91	59	49	83	39	30	74	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	28.1	30.0	31.2	33.0	35.9	36.2	36.2	36.2	35.1	33.0	31.0	30.1		
		0.4% MCWB	21.0	21.2	22.0	22.3	23.4	24.8	25.8	26.2	25.5	24.6	23.4	22.6		
		2% DB	27.3	27.9	30.0	32.1	34.0	35.1	35.0	35.1	33.9	32.4	29.1	28.0		
		2% MCWB	20.4	20.6	20.6	21.7	23.2	25.0	25.8	26.1	25.3	24.3	22.1	21.7		
		5% DB	26.0	27.3	28.0	31.0	32.8	33.9	33.9	34.0	32.9	31.3	27.8	27.3		
		5% MCWB	19.6	20.2	20.2	21.6	22.9	25.1	25.9	26.1	25.1	23.8	21.0	21.2		
		10% DB	23.7	25.9	27.3	29.8	32.2	32.7	32.8	32.8	32.4	30.2	27.0	25.9		
		10% MCWB	18.7	19.3	19.7	20.8	22.8	25.0	25.9	26.0	25.1	23.2	20.3	20.4		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	23.0	23.3	25.2	24.8	26.4	27.4	28.1	28.0	27.4	26.8	25.0	24.0		
		0.4% MCDB	26.5	26.8	27.6	29.7	31.7	32.5	32.6	33.4	31.6	30.5	28.8	27.8		
		2% WB	21.8	22.4	23.3	24.0	25.5	26.8	27.3	27.4	26.7	25.9	23.5	23.1		
		2% MCDB	25.2	26.1	26.5	28.9	30.8	32.0	32.0	32.2	31.4	29.8	27.4	26.6		
		5% WB	20.7	21.4	22.3	23.2	24.7	26.3	26.8	27.0	26.2	25.3	22.5	22.3		
		5% MCDB	23.9	25.1	25.9	27.9	30.1	31.5	31.6	31.9	31.0	29.2	26.0	25.7		
		10% WB	19.5	20.6	21.1	22.4	24.1	25.8	26.3	26.6	25.7	24.5	21.4	21.3		
		10% MCDB	23.3	24.3	24.8	27.3	29.3	30.8	31.2	31.5	30.4	28.6	24.7	24.4		
(35)	Mean Daily Temperature Range	MDBR	13.4	13.3	13.7	14.1	13.3	10.9	10.2	10.0	10.3	12.0	13.3	12.6		
		5% DB MCDBR	13.8	13.9	14.2	14.9	14.5	12.4	11.4	11.1	11.0	11.9	13.2	12.5		
		5% DB MCWBR	7.5	7.1	6.4	6.0	5.4	4.5	4.2	4.2	4.2	4.9	6.7	6.5		
		5% WB MCDBR	11.7	11.6	10.9	12.3	12.2	10.4	10.1	10.1	9.7	9.9	10.9	10.6		
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.354	0.367	0.389	0.425	0.451	0.475	0.456	0.421	0.389	0.354	0.340		
		taud	2.517	2.492	2.442	2.346	2.304	2.309	2.244	2.314	2.383	2.451	2.496	2.547		
		Ebn at Noon	903	919	924	907	870	842	821	836	858	869	881	888		
		Edh at Noon	91	101	112	127	133	131	140	129	117	104	92	85		
(44)	All-Sky Solar Radiation	RadAvg	3.53	4.24	5.36	6.41	6.76	6.08	5.79	5.58	5.09	4.69	3.83	3.26		
		RadStd	0.25	0.27	0.43	0.32	0.39	0.49	0.36	0.35	0.32	0.30	0.20	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	-4	-45	+158	+111

Nomenclature: See separate page